

WASP-36b

R-0821

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_231206 · created 2026-07-28T14:42:20+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460284.9297 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.048
Transit depth (Rp/Rs)^2	2.74 +/- 1.59 [%]
Orbital Inclination (inc)	80.8 +/- 2.92
Airmass coefficient 1 (a1)	0.78 +/- 0.44
Airmass coefficient 2 (a2)	0.1 +/- 0.72
Scatter in the residuals of the lightcurve fit is	50.28 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	6.75
Transit Duration (day)	0.05 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.048 vs 0.1368 (deviation 0.61σ)
Duration fit vs published	0.05 d vs 0.0758 d (deviation 0.81σ)
Mid-time O–C	-12.2 min
Depth SNR	3.5
Residual scatter	50.28 %

Light curve

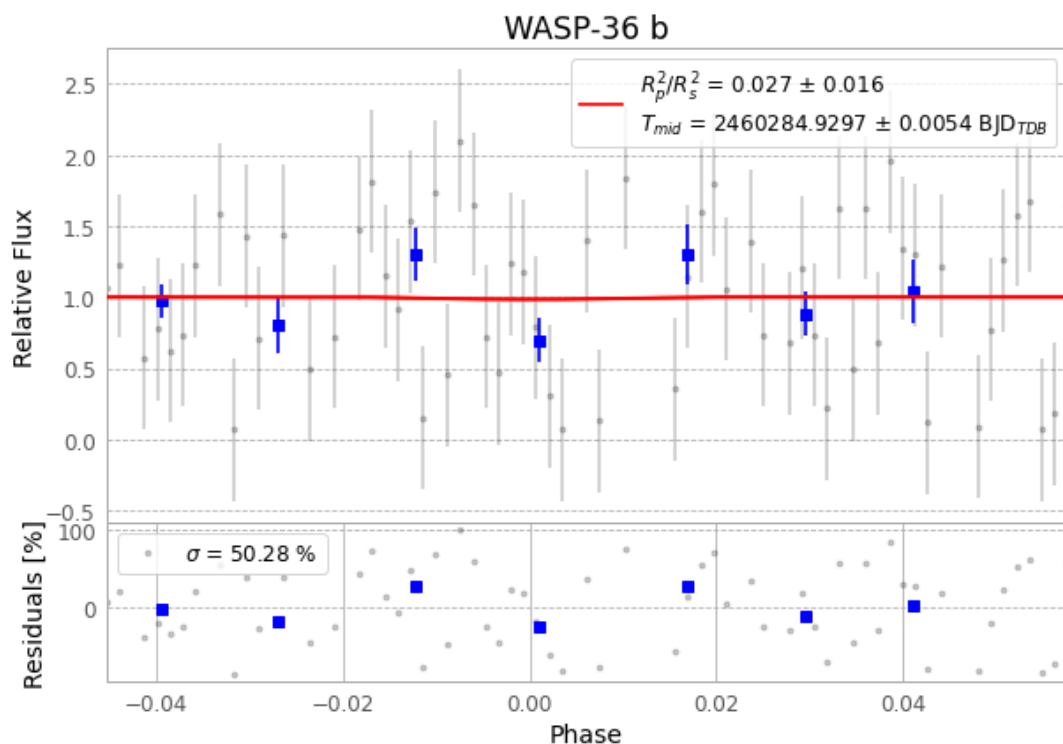


Figure 1 — FinalLightCurve_WASP-36 b_2023-12-06.png (SHA-256 1269bd029f7abee0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [549, 400], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d22f4fa8bbac4583...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36231206083315.FITS → WASP-36231206122115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-231206.log (18638bf9294f...), FinalLightCurve_WASP-36 b_2023-12-06.png (1269bd029f7a...), FinalLightCurve_WASP-36 b_2023-12-06.csv (e7201e5516a1...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 14:42Z.